Class: TE **Subject: AISC** **Sem-V**

Experiment No: 6

Course Outcome: LO3 **Blooms Level: L3**

Aim: Implement a Simple Kohonen Self-Organizing Map

Abstract: This experiment demonstrates a simple implementation of the Kohonen Self-Organizing Map (SOM) for unsupervised learning. The program uses two neurons with initially assigned weights and processes normalized student marks as input patterns. For each input, the distance between the input and the weights of both neurons is calculated, and the neuron with the minimum distance is selected as the winner neuron. The winner's weight is then updated using the learning rate. By repeatedly applying this process, the neurons gradually organize themselves according to the input patterns. The experiment helps to understand competitive learning, winner selection, and weight adaptation without using a predefined target output.

Sample Input and Output:

Sample Input

The program uses the following predefined input data:

Input Data:
0.35, 0.40, 0.45, 0.75, 0.80, 0.90

Initial weights:

Neuron 1 = 0.30
Neuron 2 = 0.80

Learning rate: *0.5*

Sample Output

Initial Weights: [0.3 0.8]

Input : 0.35
Winner Neuron : 1
Updated Weights : [0.325 0.8]

Input : 0.4
Winner Neuron : 1
Updated Weights : [0.3625 0.8]

Input : 0.45
Winner Neuron : 1
Updated Weights : [0.40625 0.8]

Input : 0.75
Winner Neuron : 2

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Updated Weights : [0.40625 0.775]

Input : 0.8
Winner Neuron : 2
Updated Weights : [0.40625 0.7875]

Input : 0.9
Winner Neuron : 2
Updated Weights : [0.40625 0.84375]

Final Weights : [0.40625 0.84375]
```

Theory:

A **Kohonen Self-Organizing Map (SOM)** is an **unsupervised neural network** developed by **Teuvo Kohonen**. It is used to organize input patterns into groups based on their similarity. In supervised learning, the network is given:

Input + Desired Output

But in SOM, only the input is provided:

Input
↓
Network identifies similar patterns

Therefore, SOM does not require predefined class labels.

Suppose we give the following student marks:

35, 40, 45, 75, 80, 90

We can expect the lower marks to be similar to each other and the higher marks to be similar to each other.

The SOM tries to **organize the neurons automatically** according to these similarities.

In our program, there are two neurons:

Neuron 1 → Initial weight = 0.3
Neuron 2 → Initial weight = 0.8

The normalized input values are: 0.35, 0.40, 0.45, 0.75, 0.80, 0.90

For every input xx, the distance from each neuron is calculated. Since there is only one input feature, the distance is:

d=|x−w|d = |x-w|

where:

- xx = input value
- ww = neuron weight

The neuron having the **smallest distance** becomes the **winner neuron**. This winner selection is also called **Best Matching Unit (BMU)** in SOM terminology.

After selecting the winner, its weight is updated using:

$$w_{new}=w_{old}+\alpha(x-w_{old})$$

where:

- w_{new} = new weight
- w_{old} = old weight
- x = input
- α = learning rate

In this experiment: α=0.5

Consider the first input: x=0.35

Initial weights: w1=0.30 w2=0.80

Calculate distances

For Neuron 1: d1=|0.35−0.30|=0.05

For Neuron 2: d2=|0.35−0.80|=0.45

Since: 0.05<0.45

Neuron 1 wins.

Update Neuron 1

w_{new}=0.30+0.5(0.35−0.30)=0.325

Therefore: Neuron 1 = 0.325 Neuron 2 = 0.800

The winning neuron has moved closer to the input value 0.35.

Now: $x=0.40$

Current weights: $w1 = 0.325$ $w2 = 0.800$

Distances: $d1=|0.40-0.325|=0.075$ $d2=|0.40-0.80|=0.40$

Neuron 1 wins again.

Update:

$w1=0.325+0.5(0.40-0.325)=0.3625$

Thus:

$w1 = 0.3625$
$w2 = 0.8$

The first three inputs are:

0.35
0.40
0.45

They are closer to Neuron 1, so Neuron 1 keeps winning and its weight moves toward these values.

The next inputs are:

0.75
0.80
0.90

These are closer to Neuron 2, so Neuron 2 wins and adapts toward the higher values.

Finally, the weights become approximately:

Neuron 1 → 0.40625

Neuron 2 → 0.84375

This shows how the neurons **self-organize according to the input data**.

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Algorithm

Simple Kohonen SOM

Step 1: Initialize the weights of the neurons.

W1 = 0.3 W2 = 0.8

Step 2: Set the learning rate.

$\alpha = 0.5$

Step 3: Take one input pattern at a time.

Step 4: Calculate the distance between the input and each neuron's weight.

$di = |x - wi|$

Step 5: Select the neuron having the minimum distance as the **winner neuron**.

Step 6: Update the winner's weight using:

$wnew = wold + \alpha(x - wold)$

Step 7: Repeat Steps 3–6 for all input patterns.

Step 8: Display the final weights.

Suppose the inputs represent normalized student marks.

35, 40, 45, 75, 80, 90

The network has two neurons.

After learning:

Neuron 1 → Lower-mark group

Neuron 2 → Higher-mark group

The important point is that **we never tell the network which student belongs to which group**. The grouping emerges from the similarity of the input values.

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Code:

https://colab.research.google.com/drive/1nNjOFMyNA13yVGrOHa8NLBMwRKtrt__t?usp=sharing

Conclusion:

The experiment demonstrates the basic working of a Kohonen Self-Organizing Map using unsupervised competitive learning. For each input pattern, the neuron closest to the input is selected as the winner, and its weight is adjusted toward the input using the learning rate. The final weights show that the neurons have gradually adapted to different regions of the input data. Thus, the experiment provides an understanding of winner selection, competitive learning, and self-organization.

Exercise 1:

Modify the program by changing the input data to: ***0.10, 0.15, 0.20, 0.70, 0.75, 0.85***

Using the same initial weights and learning rate, determine:

- The winner neuron for each input.
- The updated weights after each input.
- The final weights.

Exercise 2:

Modify the program to use **three neurons** instead of two, with initial weights: ***[0.2, 0.5, 0.8]***

Use the following input data: ***0.15, 0.25, 0.45, 0.55, 0.75, 0.90***

Use a learning rate of **0.5** and identify the winner neuron for each input.